

# Ground Station API

## Method

### Query devices

#### Description:

List all devices.

#### Example:

GET /api/devices

#### Response:

| Parameter | Description |
|-----------|-------------|
| id        | Device's id |

#### Example:

```
{
  "devices": [
    {
      "id": "RCKH"
    },
    {
      "id": "TPE"
    }
  ]
}
```

### Query a device

#### Description:

List device by device id.

#### Request:

| Parameter | Description | Type   |
|-----------|-------------|--------|
| Device_id | Device id   | String |

#### Example:

GET /api/devices/RCKH

**Response:**

| Parameter | Description   |
|-----------|---------------|
| Id        | Device's id   |
| Name      | Device's name |

**Example:**

```
{
  "RCKH": [
    {
      "id": "RCKH",
      "name": "CONSOLE40-RCKH"
    }
  ]
}
```

## Create a device

**Description:**

Create device by device id.

**Request:**

| Parameter | Description | Type   |
|-----------|-------------|--------|
| Device_id | Device id   | String |

**Example:**

POST /api/devices/RCKH

**Response:**

No specific response. It returns an empty success response if completed without error

## Query sessions

**Description:**

List all sessions.

**Example:**

GET /api/sessions

**Response:**

| Parameter | Description |
|-----------|-------------|
| id        | Session id  |

### Example:

```
{
  "sessions": [
    {
      "id": 3,
      "startTime": "Mon, 10 Jan 2022 17:57:58 GMT"
    },
    {
      "id": 4,
      "startTime": "Fri, 14 Jan 2022 16:26:59 GMT"
    },
    {
      "id": 5,
      "startTime": "Fri, 14 Jan 2022 16:27:05 GMT"
    },
    {
      "id": 6,
      "startTime": "Fri, 14 Jan 2022 16:28:45 GMT"
    },
    {
      "id": 7,
      "startTime": "Wed, 19 Jan 2022 10:43:29 GMT"
    },
    {
      "id": 8,
      "startTime": "Wed, 26 Jan 2022 09:39:51 GMT"
    }
  ]
}
```

## Query a session

### Description:

List session details by session id.

### Request:

| Parameter  | Description | Type    |
|------------|-------------|---------|
| Session_id | Session id  | Integer |

### Example:

GET /api/sessions/8

**Response:**

| Parameter   | Description                      |
|-------------|----------------------------------|
| Id          | Session id                       |
| Name        | Session start time               |
| Device list | List all devices in this session |

**Example:**

```
{
  "session": {
    "devices": [
      {
        "id": "RCKH",
        "name": "CONSOLE40-RCKH"
      },
      {
        "id": "RCKH",
        "name": "CONSOLE40-RCKH"
      }
    ],
    "id": 8,
    "startTime": "Wed, 26 Jan 2022 09:39:51 GMT"
  }
}
```

## Create a session

**Description:**

Create session.

**Example:**

POST /api/sessions

**Response:**

| Parameter  | Description |
|------------|-------------|
| Session_id | Session id  |

**Example:**

```
{
  "session_id": 78
}
```

## Add device

### Description:

Add device to session.

### Request:

| Parameter  | Description       | Type    |
|------------|-------------------|---------|
| Session_id | Session id number | Integer |
| Device_id  | Device id         | String  |

### Example:

```
POST /api/sessions/78/XXXX
```

### Response:

No specific response. It returns an empty success response if completed without error

## Start session

### Description:

Start a session for record flight data.

### Request:

| Parameter  | Description | Type    |
|------------|-------------|---------|
| Session_id | Session id  | Integer |

### Example:

```
POST /api/start_session/78
```

### Response:

No specific response. It returns an empty success response if completed without error

## Stop session

### Description:

Stop a session.

### Request:

| Parameter  | Description | Type    |
|------------|-------------|---------|
| Session_id | Session id  | Integer |

### Example:

```
POST /api/stop_session/78
```

### Response:

No specific response. It returns an empty success response if completed without error

# Query flight records

## Description:

List a device flight records in a session.

## Request:

| Parameter  | Description       | Type    |
|------------|-------------------|---------|
| Session_id | Session id number | Integer |
| Device_id  | Device id number  | String  |

## Example:

GET /api/flight\_data/1/RCKH

## Response:

### Example:

```
{
  "RCKH ": [
    {
      "aileronAxis": 0.0,
      "airspeedSel": 0.0,
      "alternator": 0,
      "altitude": 503.0,
      "altitudeSel": 3000.146480102539,
      "aoa": 0.0,
      "autopilot": 0,
      "avionics": 0,
      "battery": 1,
      "beacon": 0,
      "device_id": "RCKH",
      "elevatorAxis": 0.0,
      "elevatorTrim": -16.070556640625,
      "flaps": 1,
      "flightDirector": 0,
      "flightDirectorBank": 0.0,
      "flightDirectorPitch": 0.0,
      "fuelPump": 1,
      "fuelSelector": 2,
      "fuelWeight": 127.89115646258503,
      "heading": 106.25099387951195,
      "headingSel": 89.000244140625,
```

```
"ias": 0.0,  
"id": 1,  
"landing": 0,  
"landingGear": 1,  
"latitude": 22.33406666665975,  
"longitude": 114.1735194444449,  
"mach": 0.0,  
"magnetos": 4,  
"maxRPM": 0.9797940254211426,  
"mixtureLever": 100.0,  
"navigation": 0,  
"overspeed": 0,  
"parkingBrake": 0,  
"pitch": -2.4742944817990065,  
"pitot": 1,  
"pressure": 1013.1875,  
"propellerLever": 100.0,  
"roll": 0.8955518715083599,  
"rpm": 0.0,  
"session_id": 1,  
"sideSlip": 0.0,  
"slipSkid": 0.0,  
"stall": 0,  
"status": "pausing",  
"strobes": 0,  
"taxi": 0,  
"throttleLever": 58.59375,  
"time": "Fri, 07 Jan 2022 11:35:44 GMT",  
"turnRate": 0.0,  
"verticalSpeed": -417.538153125,  
"whiskeyCompass": 0.0
```

```
},
```

```
]
```

```
}
```

## Query session flight records

### Description:

List all flight records in a session.

### Request:

| Parameter  | Description       | Type    |
|------------|-------------------|---------|
| Session_id | Session id number | Integer |

### Example:

GET /api/flight\_data/1

### Response:

#### Example:

Working on it